

Building an Illustrated World in *The Bad Guys*

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Figure 1: The graphic art direction required development of various stylistic techniques

ABSTRACT

The artistic style of *The Bad Guys* is inspired by the strong and simplified details of 2D illustration with its hand-drawn imperfections. Similar to traditional artists, our film needed techniques to selectively apply detail & thoughtfully deconstruct objects, focusing on artistic flexibility while maintaining scalability.

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1 PAINTERLY FILTER

Computer-generated renders provide an incredible amount of high-frequency information (eg. hair, foliage, textures, etc) which conflicts with how an artist might paint these objects (via simplified shapes, silhouettes and values). To reduce superfluous detail into painterly strokes while retaining sharpness, we developed a custom anisotropic Kuwahara filter [Kyprianidis et al. 2009], which adapts to local feature direction and object coherency.

A number of improvements were required to adapt the technique for ray-traced input. Superluminous values, spatial/temporal render noise, and fireflies (ie. erroneous pixels) all have significant impact on the computation of the structure tensor and resulting orientation and isotropy estimation. Further, the kernel weight computation needs to account for partial/zero alpha in the input, otherwise an undesired darkening occurs on the edges of objects. This allowed the option to extend the effect over object boundaries or retain edge fidelity (maintaining details such as hair fly-aways, etc).

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A uniform filtering heuristic tends to over-process the image, hence we needed a targeted approach. For artist control and flexibility, we introduced fully-customizable per-pixel weights to vary the kernel's properties (eg. size, amplitude, sharpness, shape elongation, etc). In practice, artists would art-direct these weights (via depth gradients, positional falloff, luminance bands, cryptomattes, roto-scoping) to facilitate effects such as painterly depth-of-field, shadow-side detail removal, simplifying foliage/crowd detail, etc.



Figure 2: Painterly filter coalesces values into strokes

2 HAND-DRAWN DETAILS

Stylistically, it was important to complement our environments with hand-drawn linework, as image-processing techniques do not replace an artist's visual intent, deconstruction, and spontaneity. We developed the 3D drawing tool-set *Squiggles* to apply selective details across any number of assets. The artist navigates the environment sketching 2D strokes in the viewport, and the system intelligently places the strokes in non-planar 3D space. 3D placement was critical to maintain registration during camera moves and re-usability across different camera angles in a sequence.

A *Squiggles* stroke's placement is derived via per-vertex distance minimalization along the viewing ray to the closest scene object. The stroke is not required to intersect any object, and multiple hits result in refinement (or splitting), maintaining smooth placement on depth discontinuities. Drawing is performed in real-time in screen-space and the projection logic computation is hidden from the user post-stroke and cached. Artists can paint modifications into existing strokes (positional nudging, color, thickness) and/or perform per-stroke operations (disabling/deleting, transforms, duplicating subsets of strokes, procedural smoothing, noise and texture).

Squiggles are installed into USD and reused across shots in an environment, though hero-shot adjustments are easily introduced when necessary. In downstream departments, we developed tools to perform rendertime modifications (thickness, transforms, opacity, deleting) as well as compositing viewer selection for individualized stroke grading control.



Figure 3: Scalable hand-drawn details via *Squiggles*

3 FOLIAGE AUTHORING

Foliage is typically composed of many thousands of leaves (or grass, etc), yielding a visually-complex result. Image-processing was not enough to obtain the desired brushed stylistic effect and we needed to re-imagine how assets were constructed.

For assets with instanced leaves, leaf-based material shaders do not have spatial coherence when instanced. To that end, we established a tree-space texture coordinate system to attach impasto-style albedo maps. For a spatially-coherent light-response, we introduced a smooth normal [Peterson and Lee 2006], computed via filtered gradient of the canopy level set, stored & sampled as an OpenVDB vector grid.

For graphic grasses and short plants, we utilized *Sprinkles*, our internal tool-set for painting instanced geometry interactively with easily-customizable brushes. We extended the tool-set to automatically ingest 2D illustrations of plants, which allowed us to focus on silhouettes and negative space, as well as simplify internal detail.

Sprinkles combines the interactivity and stylization of a paintbrush and the power of procedural distribution, including per-instance scale, alignment, asset frequency, and shader variation. Previously-authored instances can be modified via additional strokes (color, scale, orientation, culling, etc) and/or users can select and tweak any instance with manual operations, including transformation, deletion, and duplication.

For improved integration, instances inherit shader properties from the underlying geometry (blended with per-instance variation or painted color). Transparency is adjusted 1) near the ground intersection to further integrate the sprites into the environment and 2) to feather the tips of our plant brushes. *Sprinkles* are highly-scalable, installed as USD assets for reuse across cameras in a sequence and are easily editable at rendertime & compositing.

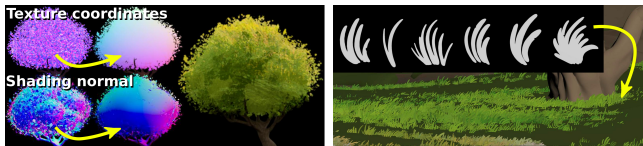


Figure 4: Graphic foliage via spatially-coherent shading techniques (left) and instanced illustrations (right)

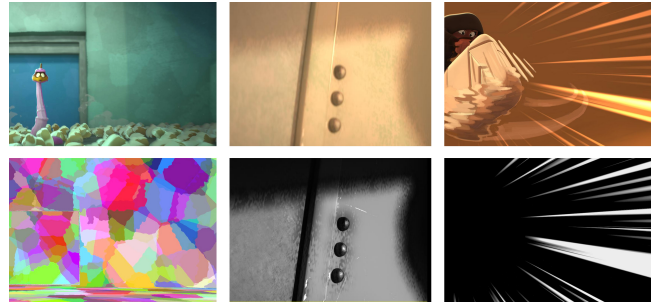


Figure 5: Various texture and linework techniques to enhance the film's style

4 LIGHTING & COMPOSITING

We developed a number of techniques to further enhance the image with painterly effects and artistic imperfections. Artists often isolated image component AOVs and modulated them via texture in compositing. The artist could perform hue shifts in the albedo and/or luminance of a light component. A library of brush textures could be assigned via tilable UDIM or interactive placement for reference space projection, each supporting motion & deformation of the underlying geometry. The technique supports projection onto difficult geometry such as hair/fur/foliage, and position/normal information can be filtered for the desired spatial coherence.

Spotches, akin to slight differences in the application of brush strokes, were achieved via perturbing the image by successive layers of cellular noise where the input positional space is first deformed via high-frequency fractal noise. The effect could be refined where voronoi cell centers act as individually-unique texture projection coordinate spaces. These pattern breakup techniques were further developed to distort shadow terminators to give the appearance of shadows with brushstroke-feathered edges.

To accentuate details in areas of low-contrast, we developed procedural line techniques that analyzed the input beauty, normal, and positional image data. Artists would scope these lines via geometry orientation about a directional light and dial in the desired line properties interactively.

True reflections were often too visually-complex to integrate in our graphic world. To provide a clean read in these situations, artists replaced in-render geometry with simpler primitives or would design/place graphic shapes interactively in the compositing viewer.

To heighten suspense in high-action sequences, we introduced the traditional 2D animation technique of *Action lines*. These graphic intrusions also complement fast camera moves & zooms, existing either orthogonal to the camera or aligned to match the perspective transform. Artists direct spatial & temporal line choreography interactively, as well as textural quality and how the beauty color is inherited & extruded.

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